

10-Week Development Plan

Project: RPS Arena – Rock Paper Scissors Game

Week 1: Project Research & Requirement Analysis

- Understand the rules of Rock-Paper-Scissors.
- Analyze project requirements and expected outcomes.
- Define game features and objectives.
- Prepare initial project documentation.

Week 2: Project Design & Planning

- Design the overall game structure.
- Create flowcharts and basic game logic diagrams.
- Plan user interaction and menu options.
- Finalize project architecture.

Week 3: Development Environment Setup

- Install and configure Java JDK and IDE.
- Create the project structure.
- Set up source files and package organization.
- Test the development environment.

Week 4: User Interface Development

- Develop a console-based user interface.
- Create menus and user input options.
- Implement input validation.
- Display game instructions and rules.

Week 5: Core Game Logic Implementation- Implement Rock, Paper, and Scissors choices.

- Develop winner determination logic.
- Handle tie situations.
- Test different game scenarios.

Week 6: Computer Opponent Development

- Implement random choice generation for the computer.
- Integrate computer decisions with the game logic.
- Verify fairness of random selection.
- Conduct functionality testing.

Week 7: Score Tracking System

- Add score counting functionality.
- Track player wins, computer wins, and draws.
- Display score after each round.
- Test score calculations.

Week 8: Additional Features & Enhancements

- Add multiple-round gameplay.
- Implement replay functionality.
- Improve user experience and interface messages.
- Handle unexpected user inputs.

Week 9: Testing & Debugging

- Perform comprehensive testing.
- Fix identified bugs and logical errors.
- Optimize code readability and performance.
- Validate all project requirements.

Week 10: Documentation & Final Submission

- Prepare final project report.
- Create project presentation slides.
- Demonstrate project functionality.
- Submit the completed project.